Exp. No: 20

PERFORMANCE EVALUATION OF FREE-SPACE OPTICAL (FSO) LINK UNDER ATMOSPHERIC TURBULENCE USING OPTISYSTEM

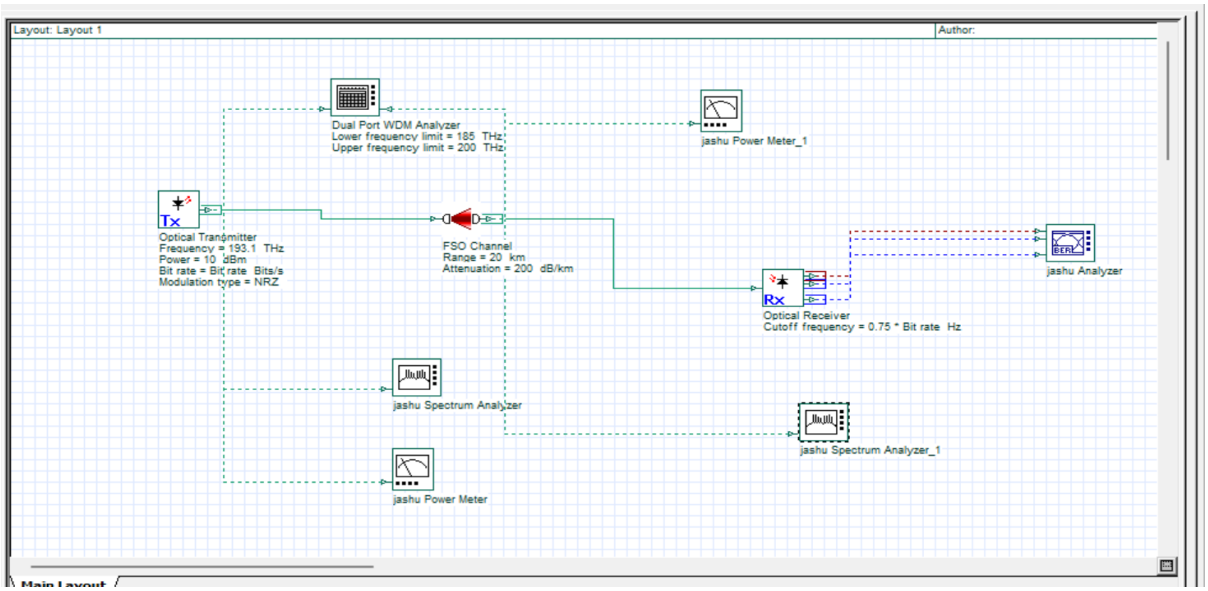
OBJECTIVES

1. Analyze the effect of range on received power, considering the combined contributions of geometric spreading loss and atmospheric attenuation.
2. Evaluate how the combined effects of increasing range and attenuation influence received power and bit error rate (BER).
3. Assess the extent to which increased transmit power can compensate for attenuation-induced signal loss.

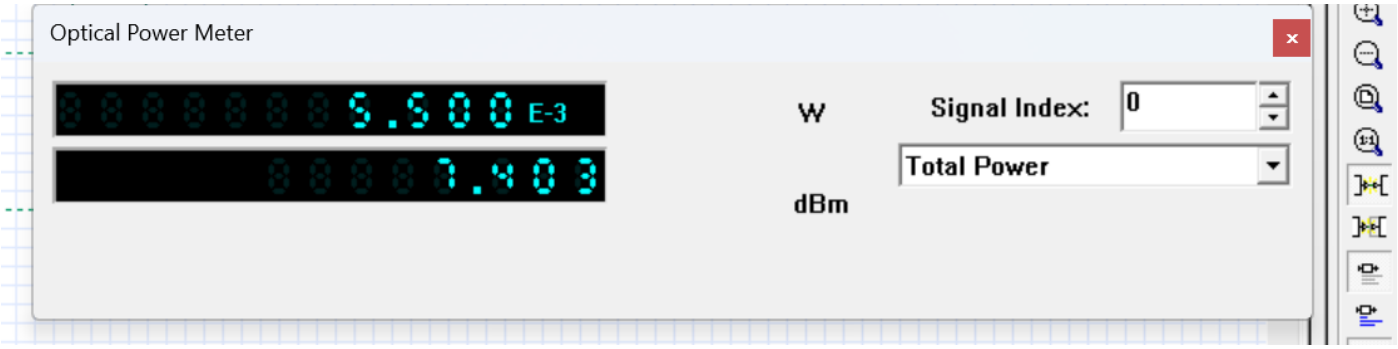
SOFTWARE REQUIRED

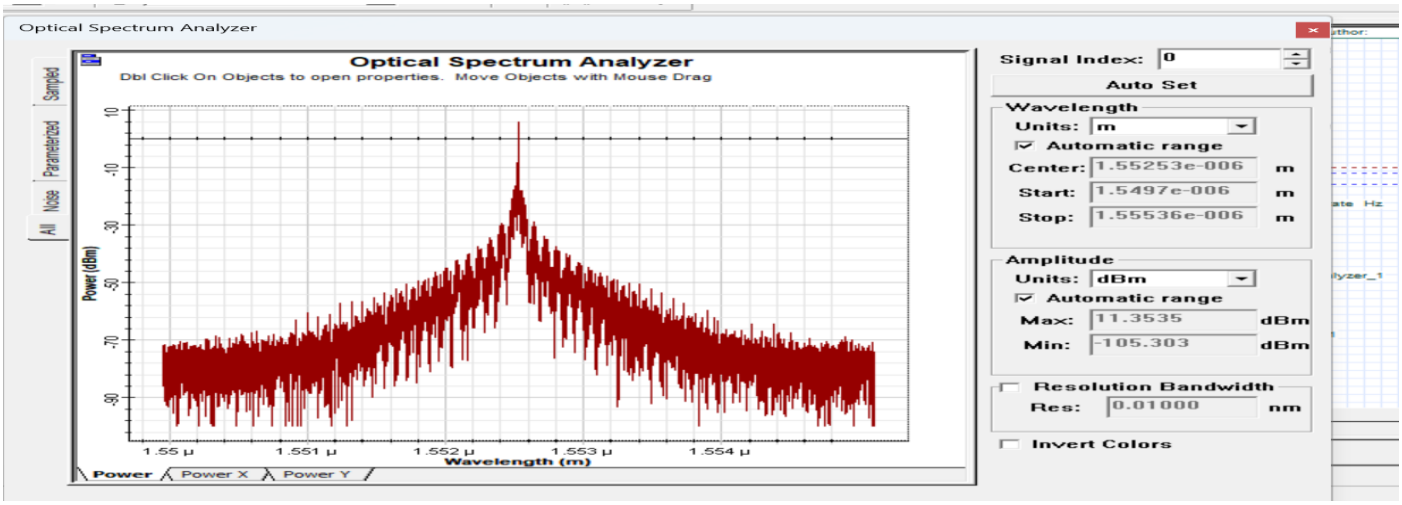
- Optisystem

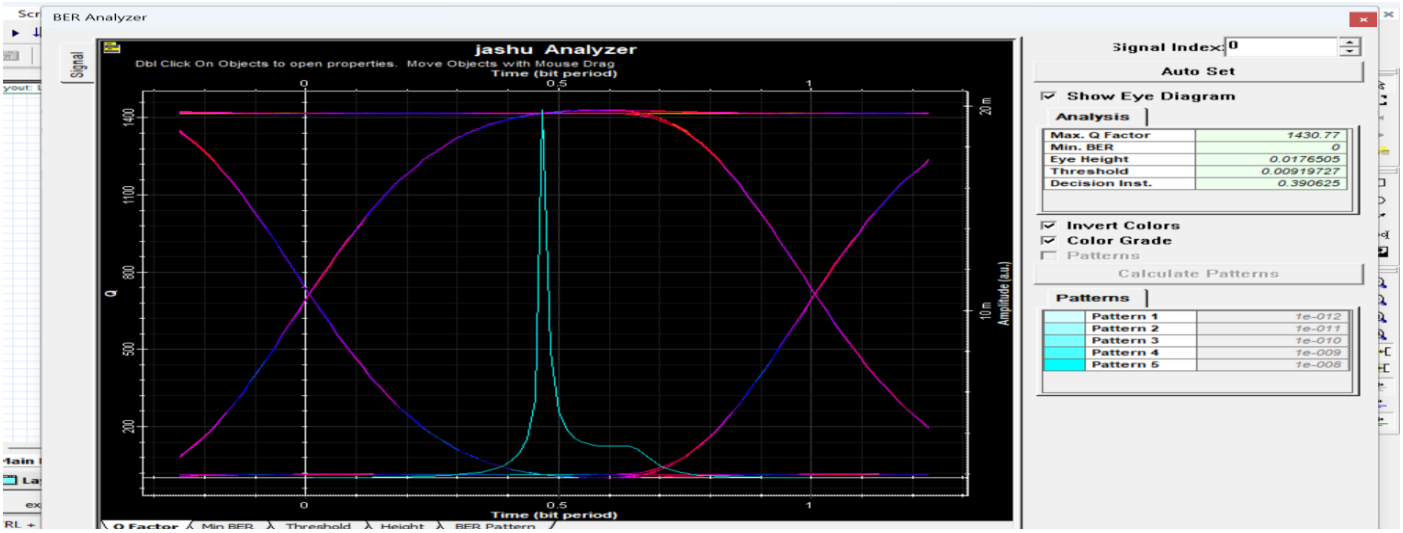
BLOCK DIAGRAM

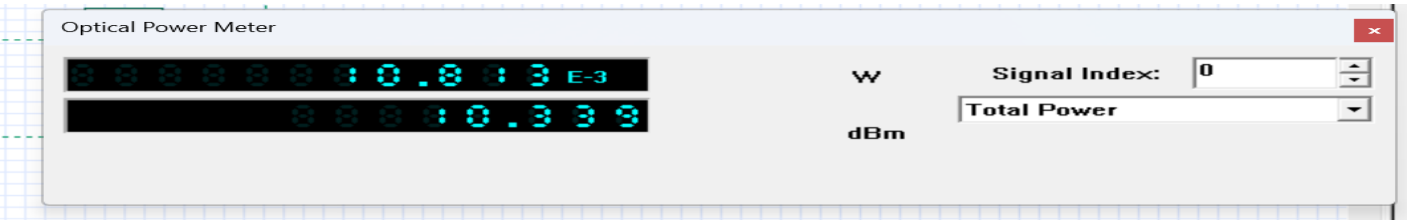


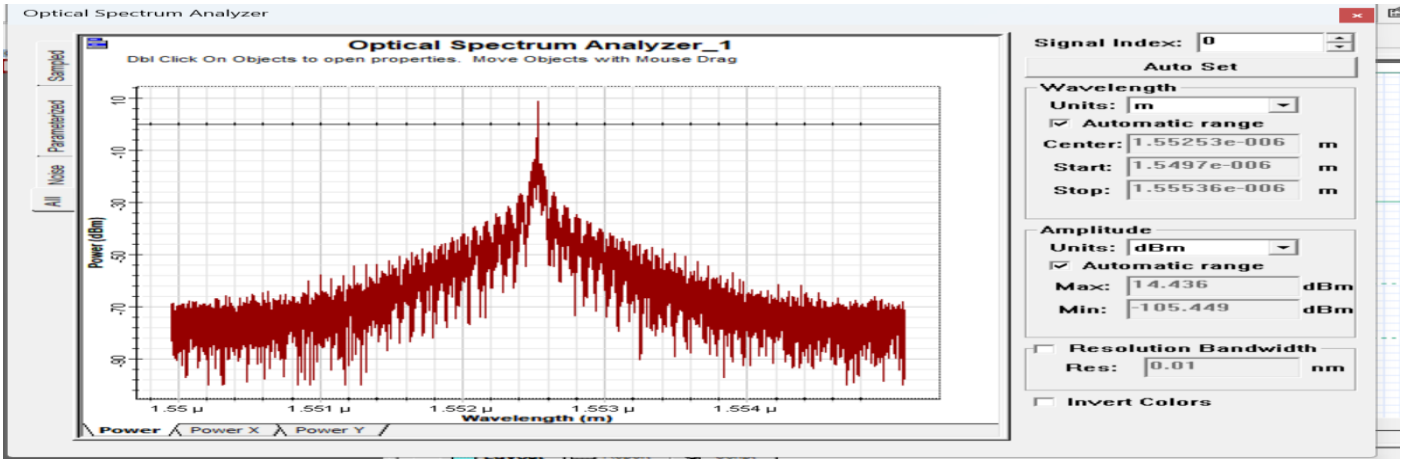
OUTPUT WAVEFORMS











Reflection: I learned how atmospheric turbulence affects free-space optical communication. I understood how weather conditions can influence signal strength and link performance.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighta ge
System Design & FSO Link Setup	Correct selection and connection of OptiSystem components for Free Space Optics link (Optical Transmitter, FSO Channel, PIN Photodetector, Low Pass Filter, Optical Power Meter, Time Domain Visualizer, BER Analyzer).	All optical transmitter, free space channel, receiver, and visualizer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the FSO link remains functional.	Component selection is partially complete; minor connection or configuratio n errors impede optical signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configurati on & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Modulation (OOK/NRZ), Transmit Power, FSO Range (1 to 15 km), and Atmospheric Attenuation (2 to 10 dB/km).	All input parameters and test cases across varying ranges and attenuation values are accurately set and executed.	Test case parameters are mostly accurate with minor setting discrepancie s.	Basic parameters are configured, but multiple range or attenuation test cases are missing.	Incorrect input parameter settings or failure to run the required atmospheric test cases.	20%
Measureme nt & Power Monitoring	Accurate extraction and recording of received optical power, Q-Factor, BER, Eye Height, and Threshold across all range settings.	Precise data extraction across all FSO range cases, neatly tabulated with complete optical power readings and clear time-domain waveforms.	Data is collected with minor omissions in recording power meter readings, BER, or eye parameters.	Partial data collection with incomplete tables or missing optical power visualizer readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Atmospheri c & Link Loss Analysis	Detailed evaluation of geometric spreading loss, atmospheric attenuation, distance-dependent signal degradation, and power compensation feasibility.	Comprehensive analysis correctly relating range and attenuation variations to geometric loss, received power drop, eye height reduction, and bit error rate performance.	Clear analysis identifying basic trends in received power and eye height relative to FSO distance.	Basic attempt at result analysis without explaining physical mechanisms like atmospheric scattering or geometric loss.	Erroneous or missing analysis of simulation results and power meter outputs.	20%
Report & Conclusion	Identification of maximum reliable communication distance, power compensation trade-offs with diminishing returns, and structural clarity.	Report features clear time-domain waveforms, complete power tables, and insightful conclusions identifying optimal operational parameters under atmospheric impairments.	Satisfactory report with relevant visualizer displays and sound concluding summary.	Incomplete report with missing graphs, power readings, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

Result: The performance of an FSO link under atmospheric turbulence was simulated and analyzed using OptiSystem.